MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY
© 1/ TYPES 7AQP7 AND 7AQP33

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Magnetic deflection, electrostatic focus

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb2	Ehk	Rg1	Eb1
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	Vdc
Maximum:	6.9	0, -180	770	12, 100	±180	Note 1	1, 100
Minimum:	5.7	---	---	7, 000	---	---	-550
TEST CONDITIONS:	6.3	Adjust	300	10, 000	---	---	Focus

GENERAL:

Qualification - Required

1/ See note 6

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
(C) 5206	Gas ratio	4		Gr	---	0.25	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5223	Modulation	-	Ib2 = 100 µAdc	ΔEc1	---	32	Vdc
5231	Spot position (magnetic deflection)	-		---	---	0.375	inch
5241	Grid cutoff voltage	-		Ec1	-15	-45	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
(C) 5246	Focusing electrode voltage	-	Ib2 = 100 µAdc	Eb1	-50	350	Vdc
1301	Heater current	-		If	270	330	mA
5101	Side terminal and base alinement	-	Halfway between pins No. 1 and No. 9	---	---	---	---
5101	Neck and bulb alinement (magnetic types)	5		---	---	---	---
5101	Face tilt	-		---	---	---	---
5101	Neck straightness	2		---	---	---	---
5216	Stray light emission (conventional types)	-	Eb2 = 12,000 Vdc; Ec2 = 770 Vdc; Eb1 = 0	---	---	---	---
5221	Screen (P7)	3		---	---	---	---
5226	Linewidth "A" (magnetic deflection)	-	Ib2 = 100 µAdc	Width	---	0.38	mm
5221	Heater-cathode leakage current	-		---	---	---	---
5251	Grid No. 2 leakage current	-		---	---	---	---
(C) 1111	Base pin solder depth	-		---	---	---	---
(C) 1101	Secureness of base, cap, or insert	-		---	---	---	---
(C) 1105	Permanence of marking	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Eb2 = 12,100 Vdc; Ec2 = 770 Vdc; Ib2 = 60 μAdc; Eb1 at focus	t	500	---	hrs
---	Life-test end points:						
5226	Linewidth "A"	-	Ib2 = 100 μAdc	Width	---	0.38	mm
5223	Modulation	-	Ib2 = 100 μAdc	ΔEc1	---	32	Vdc
	Periodic-check tests						
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		Width	---	2	mm
5201	Electrode current (grid No. 2)	-	Ec1 = 0	Ic2	-15	+15	μAdc
5216	Cathode illumination	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	Cathode to all Control grid to all	Ck Cg1	--- ---	8 10	pF pF
1216	Base material insulating quality	-		---	---	---	---

NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 Meg. When Ec2 is less than 330 Vdc, Rg1 shall not exceed 1.5 Meg.
2. Cylinder shall be 4 inches (101.6 mm) long and 0.901 inch (22.89 mm) maximum inside diameter.
3. The screen characteristics shall be measured with a constant beam energy of 0.24 watt defocused to a spot approximately 0.25 cm in diameter.
   The test conditions shall be Eb2 = 10,000 Vdc,
   Ib2 = 24 μAdc.
   The screen characteristics shall be cB5 - 450 cB minimum,
   G5:1 = 4 minimum.
4. This test to be performed at the conclusion of the holding period.
5. The two sets of neck supports referenced in method 5101 shall be spaced 1.125 inches (28.58 mm) and 3.25 inches (82.55 mm) below reference plane.
6. Tube type 7AQP33 has been added to this tube specification sheet.

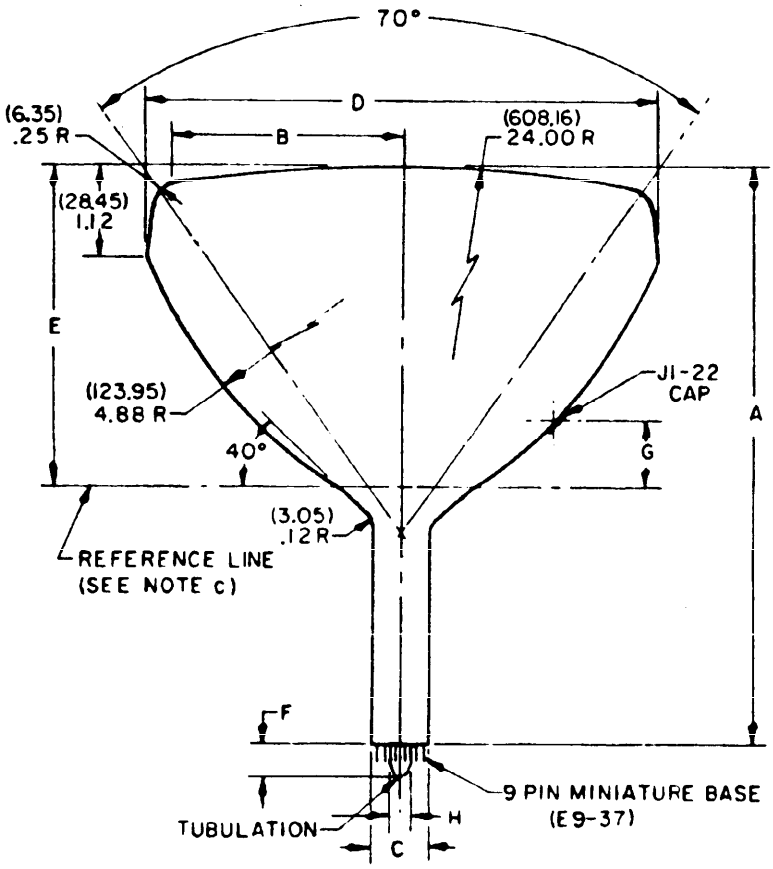
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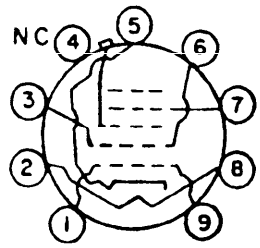
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PIN NO.	ELEMENT
1	Grid No. 1
2	Heater
3	Grid No. 2
4	No internal connection permitted
5	Cathode
6	Grid No. 2
7	Focusing electrode
8	Heater
9	Grid No. 1
CAP	Accelerator

NOTES:
a. Dimensions are in inches.
b. Metric equivalents are in parenthesis.
c. Reference line is determined by the point where leading edge of JEDEC-G-123 reference line gage will stop.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
	Quality conformance inspection, part 2	
A	7.88 (200.15)	8.25 (209.55)
B	3.00 (76.20)	- - - - - -
C	.81 (20.57)	.90 (22.86)
D	6.94 (176.28)	7.06 (179.32)
E	4.22 (107.19)	4.41 (112.01)
F	- - - - - -	.44 (11.18)
G	.84 (21.34)	1.09 (27.69)
H	- - - - - -	.25 (6.35)

FIGURE 1. Outline drawing of electron tube types 7AQP7 and 7AQP33.

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